

IoT-based Smart Parking Management System (IoT-SPMS)

Submission #1 — Requirements Specification

Course: Software Engineering (SE252) **Project:** Smart Parking System for University Campus — Ho Chi Minh City University of Technology (HCMUT) **Document status:** Requirements baseline v1.0

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1. Project Context & Overview

1.1 Background

Ho Chi Minh City University of Technology (HCMUT), a member of Vietnam National University – HCMC, operates across two campuses — **Cơ sở 1** at 268 Lý Thường Kiệt (District 10) and **Cơ sở 2** in Dĩ An, Bình Dương — serving approximately **23,000 undergraduates, 2,100 master's students, and 300 doctoral candidates**, plus faculty, staff, and external visitors. This population generates a very large volume of daily parking activity.

The university already runs an **RFID parking-card scheme** (introduced May 2019) for motorbikes and bicycles. That incumbent system, and campus parking generally, faces four concrete pain points:

- **Peak-hour congestion.** Nearly all students arrive on motorbikes in tight pre-class windows (before ~07:00 and ~12:30), producing hundreds of arrivals per minute at each gate. Per-vehicle barrier gates physically cannot keep up.
- **Card loss and counterfeiting**, with weak binding between the card and the actual vehicle.
- **Manual cash handling and manual reconciliation** by guards.
- **No real-time visibility** of where spaces are free, leading to wasted search time inside lots.

1.2 The proposed system

The **IoT-based Smart Parking Management System (IoT-SPMS)** modernises campus parking by:

- Linking every parking transaction to the driver's **real HCMUT identity** via the university Single Sign-On (**HCMUT_SSO**).
- Detecting per-slot occupancy through **IoT sensors** and computing a **near-real-time** view of availability.
- Driving **electronic guidance signage** (available / nearly-full / full) at entrances and key intersections.
- Automating **fee computation and payment** through the university payment platform (**BKPay**) for members, with cash/QR handling for visitors.
- Giving operators and administrators **live visibility, configuration, and full audit** of all parking and financial activity.

1.3 Domain reality that shapes the requirements

Two facts distinguish this from a Western car-parking product and drive design decisions throughout this document:

1. **Motorbike dominance.** In Vietnam ~89% of households own a motorbike versus ~9% a car, and motorbikes are 85–90% of road traffic. On campus the primary vehicle class is the **motorbike (xe máy)**, parked in a dense ~1 m × 2 m footprint, with cars a small minority handled in a separate lane. The system's primary flow is therefore designed for **high-throughput motorbike traffic**, not one-car-per-bay barrier stalls.
2. **Burst throughput is the defining constraint.** Because a per-vehicle barrier has a ~1.5 s open/close cycle (≈40 vehicles/min ceiling per lane), it gridlocks against peak demand. The system is designed for **barrier-free, multi-lane flow**: an overhead ANPR camera plus a contactless RFID/NFC card tap recorded on the move, with plate-versus-card reconciliation at exit. Physical barriers are reserved for the smaller car lot and paid-exit reconciliation.

A **core anti-theft rule** follows from (1)–(2): identity is **plate-bound, not merely card-bound**. On entry the system records the card/account, the ANPR-read plate, an entry photo, and a timestamp; on exit it re-reads the plate for the same card and **raises an anti-theft alarm on any mismatch** (a stolen bike presents a card that does not match its plate). This rule is reflected in the requirements below (FR-EXT-04) and in the data model.

2. Stakeholders

#	Stakeholder	Interest / Expectation	Influence
S1	Drivers who are members (students, faculty, staff)	Fast, frictionless entry/exit; fair and transparent fees; ability to see free spaces before entering; self-service account and payment.	High (primary users, largest group)
S2	Visitor drivers	Simple guest access without a campus account; clear payment at exit.	Medium
S3	Parking operators / attendants	Reliable live occupancy board; fast vehicle lookup; tools to handle exceptions (stuck barrier, lost ticket, alarms) and take cash.	High (day-to-day operation)
S4	System administrators	Configurable tariffs, zones, sensors, and roles; dashboards; exportable reports; complete audit trail.	High (own the system config)
S5	University finance department	Correct fee computation; reconciled revenue against the bank (OCB) settlement; auditable financial records.	High
S6	Facilities & campus security	Barrier safety; anti-theft protection; equipment reliability.	Medium
S7	HCMUT IT / SSO administrators	Correct, secure integration with HCMUT_SSO (CAS) and read-only integration with HCMUT_DATACORE; attribute-release governance.	High (gatekeepers of integration)
S8	University management	Reduced congestion, better space utilisation, modern student experience, cost control.	High (project sponsor)
S9	SE252 course graders	A correct, realistic, well-structured engineering deliverable.	(Academic)

3. Objectives & Scope

3.1 Objectives

- **O1 — Automate access control.** Record entry/exit automatically for members via SSO-linked identity and for visitors via a temporary-access mechanism, without a centralized-authentication dependency at the gate.
- **O2 — Provide near-real-time availability.** Maintain and publish a per-zone free-space count that is resilient to sensor and network faults.
- **O3 — Guide traffic.** Drive dynamic signage (available / nearly-full / full) and directional guidance to reduce in-lot search and gate congestion.
- **O4 — Integrate billing & payment.** Accumulate learner activity over a billing period, compute fees per configurable policy, and initiate payment through BKPay; support cash/QR for visitors.
- **O5 — Enforce roles & auditability.** Apply role-based access control across end users, operators, and administrators, and log all parking and financial activity immutably.

- **O6 — Operate reliably under real-world constraints:** high concurrency, intermittent connectivity, and heterogeneous user groups.

3.2 In scope

Access control (entry/exit), IoT slot-occupancy tracking and reconciliation, guidance signage, availability API/app, session and billing management, BKPay/cash payment, visitor ticketing, SSO authentication, read-only DATACORE synchronization, RBAC, administration/configuration, dashboards/reporting, and audit logging.

3.3 Out of scope (this project)

Physical procurement/installation of sensors and barriers; the internal implementation of HCMUT_SSO, HCMUT_DATACORE, and BKPay (treated as external systems); enforcement/towing of illegally parked vehicles; navigation *outside* the campus.

3.4 Scope decisions for the MVP

- **Reservation** (FR group E) is treated as **optional/bonus**.
- **BKPay** is integrated at the **design level** and **stubbed/mock**ed in the prototype, because the real BKPay is web-only, bank-locked (OCB), and exposes no public parking API (see §8).
- The prototype demonstrates a **single representative lot**, but the model and configuration support **multi-campus, multi-gate**.

4. Actors

Actors are separated on two axes so they are not conflated: the **application role** (what functions you may invoke — an RBAC concern) and, for end users, the **affiliation** (student / faculty / staff / visitor — an attribute that drives *pricing and eligibility*, not screens).

4.1 Primary (human) actors

Actor	Application role	Description & main goals
Member Driver	<code>end_user</code>	Student, faculty, or staff with an HCMUT identity. Authenticates via SSO-linked account, views availability, enters/exits, is billed automatically, tops up a prepaid balance, and manages own account, vehicles, and history.
Visitor Driver	<code>end_user</code> (<code>affiliation = visitor</code>)	External person with no campus account . Handled through an operator-created, plate-only/ticket session; pays at exit by cash or QR.
Parking Operator	<code>operator</code>	Monitors the live occupancy board, handles exceptions and alarms, issues visitor tickets, looks up vehicles by plate, and accepts cash. Scoped to assigned lot(s).
System Administrator	<code>admin</code>	Configures tariffs, zones, slots, sensors, devices, and roles; manages users and blacklist; views dashboards, reports, and the audit log; runs reconciliation.
Finance Officer	<code>finance</code> (optional split of admin)	Views financial reports and reconciliation breaks; authorises refunds/waivers.

4.2 Secondary (external-system) actors

These participate in use cases and appear on the use-case diagram:

- **Slot Occupancy Sensor** — reports a slot as occupied/vacant.
- **Barrier Controller** — receives open/close commands (car lane / paid exit).
- **RFID/NFC Card Reader** — reads the member card.
- **ANPR Camera** — reads the licence plate.
- **Guidance Signage** — displays availability state and directions.
- **HCMUT_SSO** — the university Central Authentication Service (CAS).
- **HCMUT_DATACORE** — read-only system of record for identity/vehicle data.
- **BKPay Payment Gateway** — processes member/online payments (via OCB).

5. Functional Requirements

Requirements are grouped into seven modules. Each is atomic and verifiable, uniquely identified, and phrased as "*The system shall...*". Priorities use **MoSCoW** (M = Must, S = Should, C = Could). Each requirement is traceable to a use case (see §6).

Module A — Entry / Access Control

ID	Priority	Requirement
FR-ENT-01	M	The system shall detect a vehicle approaching an entry lane and trigger credential capture.
FR-ENT-02	M	The system shall read a member's RFID/NFC card or QR code and read the vehicle's licence plate via the ANPR camera at entry.
FR-ENT-03	M	The system shall validate the presented credential against active membership, the blacklist, and the lot-full condition before granting entry.
FR-ENT-04	M	The system shall, when a valid member credential is present, create an active parking session recording {timestamp, lane, card/account ID, plate, entry photo}.
FR-ENT-05	M	The system shall, when no member credential is present, initiate the visitor/temporary-access flow : issue a parking ticket (QR) and open a plate-only session managed independently of HCMUT_SSO.
FR-ENT-06	M	The system shall deny entry and display a reason when the lot/zone is full, the credential is invalid, or the vehicle/plate is blacklisted.
FR-ENT-07	S	The system shall, for the car lane, command the entry barrier to open on a granted decision and record the barrier action.
FR-ENT-08	M	The system shall keep recording entry/exit during a backend or network outage using locally cached decisions, and queue the events for synchronization on reconnect (see NFR-REL-02).

Module B — Slot Occupancy & Detection

ID	Priority	Requirement
FR-OCC-01	M	The system shall receive per-slot occupancy state changes (occupied ↔ vacant) reported by slot sensors.
FR-OCC-02	M	The system shall maintain an authoritative current state for every slot: FREE, OCCUPIED, RESERVED, or OUT_OF_SERVICE.
FR-OCC-03	M	The system shall aggregate a free-space count per zone and per lot from slot state and gate entry/exit counts.
FR-OCC-04	M	The system shall reconcile the sensor-derived state against gate entry/exit counts on a periodic schedule and flag discrepancies.
FR-OCC-05	M	The system shall mark a slot/sensor whose state is older than the staleness timeout as "unknown" and shall never count an unknown slot as free (see NFR-REL-03).
FR-OCC-06	S	The system shall detect and flag a stuck or faulty sensor and raise a maintenance alert.
FR-OCC-07	S	The system shall perform a nightly zero/recalibration of count-based occupancy when a lot empties, to counter count drift.

Module C — Guidance & Signage

ID	Priority	Requirement
FR-SIG-01	M	The system shall compute an availability state per zone from the free-space count and configurable thresholds.
FR-SIG-02	M	The system shall drive an entrance sign showing the lot state (e.g. xx SPACES / NEARLY FULL / FULL).
FR-SIG-03	S	The system shall drive zone/level signs and in-aisle directional guidance pointing toward the adjacent zone with the greatest free count; when a zone is full it shall direct drivers to the nearest non-full zone.
FR-SIG-04	M	The system shall allow administrators to configure the occupancy-percentage thresholds that map to each sign state (default: green < 75 %, yellow 75–89 %, orange/nearly-full 90–99 %, red/full 100 %).
FR-SIG-05	M	The system shall publish the current availability to a web/mobile view, timestamping each figure.

Module D — Exit / Billing

ID	Priority	Requirement
FR-EXT-01	M	The system shall read the member card (or ticket QR) and re-read the licence plate at exit.
FR-EXT-02	M	The system shall locate the matching active session for the presented credential.
FR-EXT-03	M	The system shall compute the parking fee from session duration and the applicable pricing policy (vehicle type, affiliation tier, active pass), applying the configured rounding and daily-cap rules.
FR-EXT-04	M	The system shall compare the entry plate against the exit plate and raise an anti-theft alarm on mismatch, holding the session for operator resolution.
FR-EXT-05	M	The system shall process payment for the computed fee via prepaid wallet balance, BKPay, or operator-accepted cash, and close the session on success.
FR-EXT-06	M	The system shall freeze the computed amount and the pricing-rule version onto the billing record at session close, so later rate changes never rewrite historical bills.
FR-EXT-07	S	The system shall, for the car lane, command the exit barrier to open only after payment is settled or a valid pass is confirmed.
FR-EXT-08	M	The system shall accumulate member charges over the configured billing period and generate a period statement, initiating a BKPay payment request for the balance.

Module E — Reservation (optional / bonus)

ID	Priority	Requirement
FR-RES-01	C	The system shall let a member search availability and reserve a slot for a time window.
FR-RES-02	C	The system shall hold a reserved slot and auto-release it after a configurable no-show timeout.
FR-RES-03	C	The system shall convert a reservation into an active session when the member enters within the window.

Module F — Administration & Reporting

ID	Priority	Requirement
FR-ADM-01	M	The system shall let administrators create, update, version, and deactivate pricing policies/tariffs with <code>valid-from</code> / <code>valid-to</code> dates.
FR-ADM-02	M	The system shall let administrators manage users, cards, vehicles, the blacklist, and role assignments.
FR-ADM-03	M	The system shall let administrators configure zones, slots, sensors, gateways, signage, and the device registry.
FR-ADM-04	M	The system shall present dashboards for occupancy, revenue, and peak-hour analysis.
FR-ADM-05	S	The system shall export parking and financial reports for a selected period.
FR-ADM-06	M	The system shall run a periodic reconciliation of local settled payments against the bank/BKPay settlement report and record unmatched items as reconciliation breaks.

Module G — Monitoring & Audit

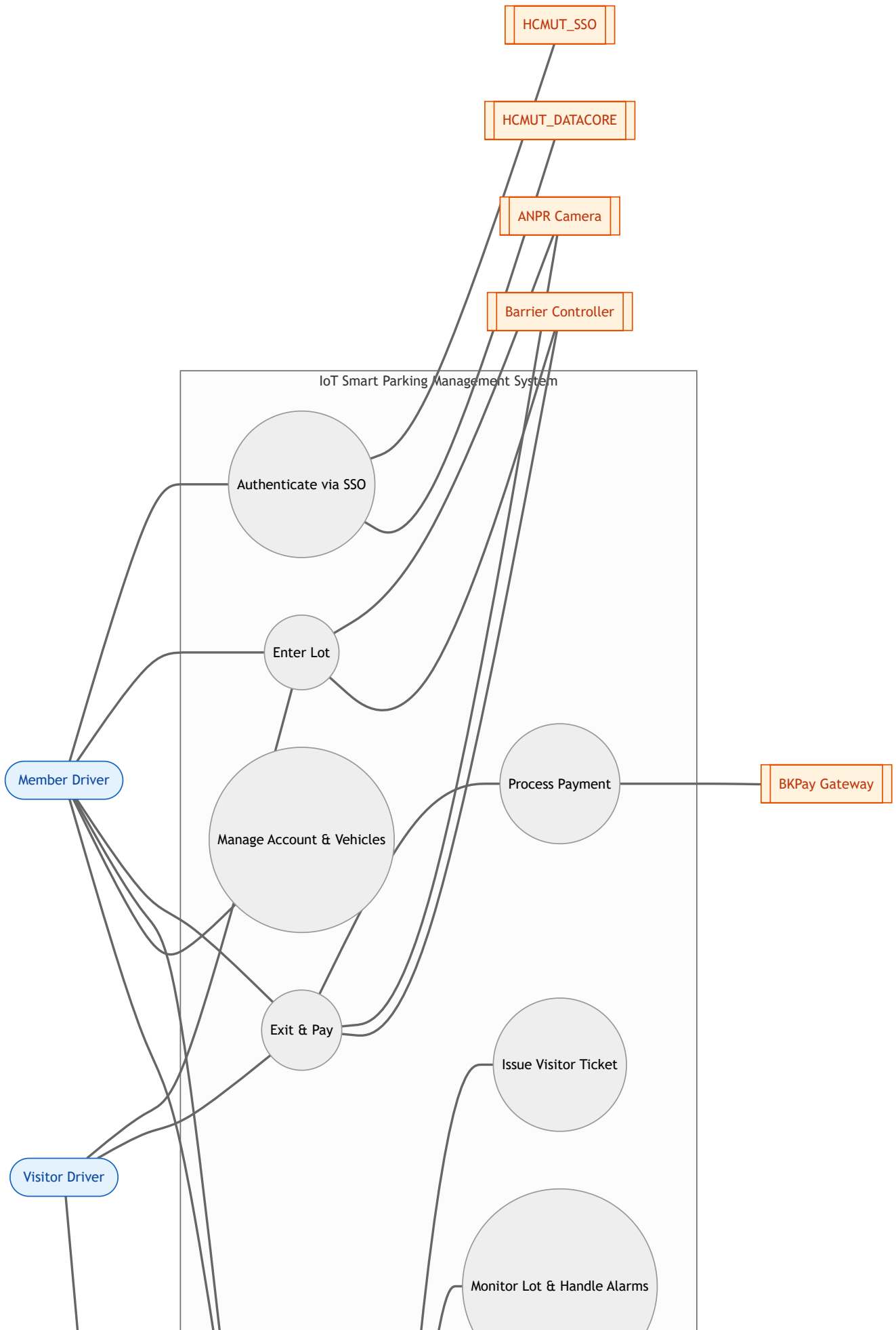
ID	Priority	Requirement
FR-AUD-01	M	The system shall present operators a live lot-status board of current occupancy, active sessions, and open alarms.
FR-AUD-02	M	The system shall let an operator look up any active session or vehicle by plate.
FR-AUD-03	M	The system shall raise alarms for hardware faults, forced barriers, plate mismatch, and anti-passback (double-entry) violations, and let operators acknowledge them.
FR-AUD-04	M	The system shall write an immutable, tamper-evident audit record for every entry, exit, payment, and privileged administrative action, capturing actor, timestamp, and before/after state.

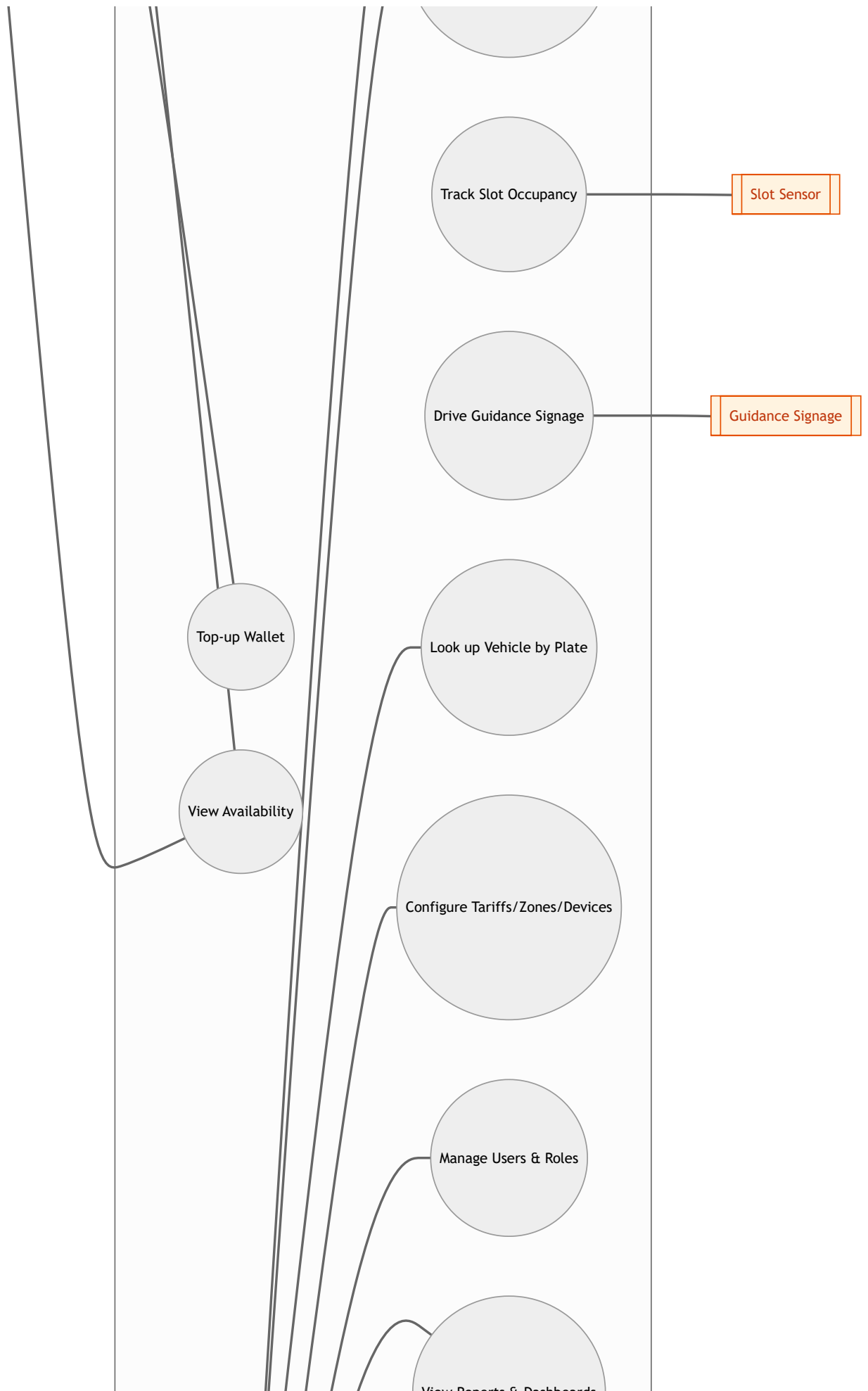
5.1 Requirement → Use-case traceability (summary)

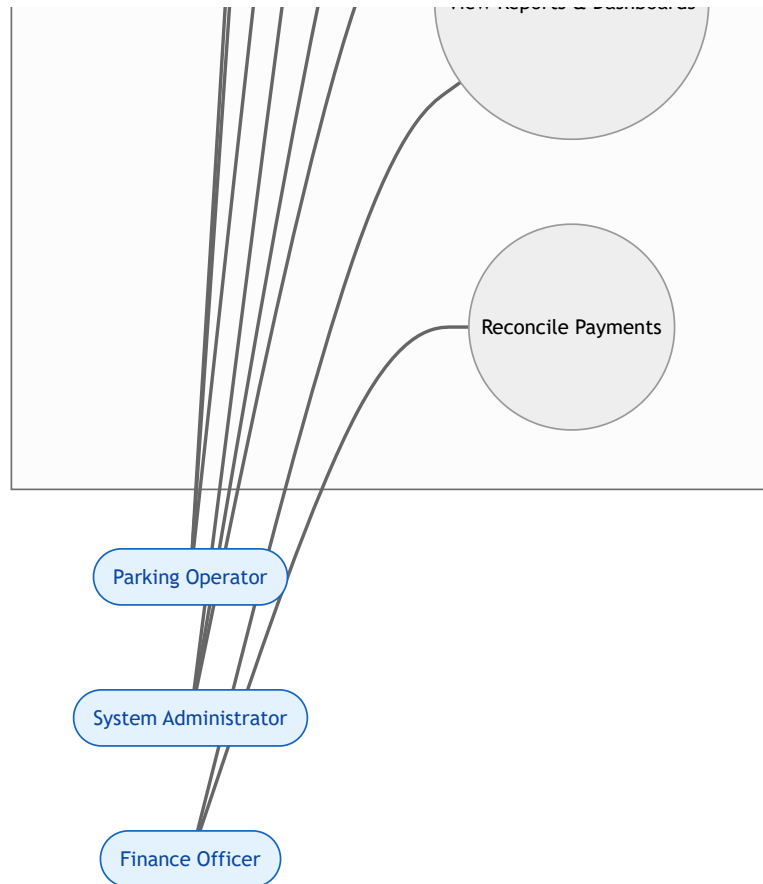
Use case	Requirements covered
Enter Lot (Member)	FR-ENT-01..04, FR-ENT-07, FR-OCC-03
Enter Lot (Visitor)	FR-ENT-01, FR-ENT-05, FR-ENT-06
Exit & Pay	FR-EXT-01..08, FR-AUD-04
View Availability	FR-OCC-03, FR-SIG-01, FR-SIG-05
Drive Signage	FR-SIG-01..04
Track Occupancy	FR-OCC-01..07
Monitor & Handle Alarms	FR-AUD-01..03, FR-EXT-04
Configure System	FR-ADM-01..03, FR-SIG-04
Reporting & Reconciliation	FR-ADM-04..06, FR-AUD-04
Authenticate (SSO)	(cross-cutting; supports all end-user use cases)

6. Use-Case Diagram

Rendered with Mermaid. Members and visitors are both `end_user` ; visitor sessions are created via the operator. External systems appear as secondary actors on the right.







Relationship notes (for the tabular UML in the PDF):

- **Enter Lot** «include» **Track Slot Occupancy** (availability check) and «include» capture via ANPR.
- **Exit & Pay** «include» **Process Payment**; «extend» *Anti-theft alarm on plate mismatch* (FR-EXT-04).
- **Enter Lot** «extend» *Issue Visitor Ticket* (when no member credential — FR-ENT-05).
- **Authenticate via SSO** is «include» d by all member-initiated use cases.
- **View Reports & Dashboards** generalises to Finance and Admin.

7. Non-Functional Requirements

Classified by **ISO/IEC 25010:2023** product-quality characteristics. Each states a **measurable target the team will validate** (not a measured result) with a verification method. Priority order for this domain: (1) offline/availability, (2) sensor fault tolerance, (3) latency, (4) concurrency, (5) security/privacy, (6) auditability, (7) scalability, (8) usability, (9) safety.

ID	25010 characteristic	Requirement	Metric / target	Condition / load	Verification
NFR-PERF-01	Performance efficiency (time behaviour)	Barrier-open latency from a valid card read to the open command.	$\leq 2 \text{ s (p95)}, \leq 3 \text{ s (p99)}$	Nominal load	Load test with timing logs
NFR-PERF-02	Performance efficiency	Availability view and signage reflect a slot state change.	$\leq 5 \text{ s}$	Near-real-time	Instrumented end-to-end test
NFR-PERF-03	Performance efficiency	Core API response time.	$\text{p99} \leq 500 \text{ ms}$	Nominal	Load test
NFR-PERF-04	Performance efficiency	ANPR plate recognition.	$< 2 \text{ s}, \geq 98 \% \text{ accuracy}$	Daylight, standard plates	Sample dataset test
NFR-CAP-01	Performance efficiency (capacity)	Concurrent lane events / active sessions / sensor ingest without degradation.	$\geq 20 \text{ lane events}; \geq 2,000 \text{ active sessions}; \geq 500 \text{ msg/s ingest}$	Peak	Stress test
NFR-REL-01	Reliability (availability)	System uptime.	$\geq 99.5 \% \text{ monthly } (\leq \sim 3.6 \text{ h/month})$	Rolling month	Uptime monitoring
NFR-REL-02	Reliability (offline operation)	Entry/exit continues during a backend/network outage; queued events sync on reconnect.	Local decisions cached; sync $\leq 60 \text{ s}$ of reconnect	Backhaul outage	Fault-injection test
NFR-REL-03	Reliability (fault tolerance)	A single failed sensor must not corrupt the free-count; slot degrades to "unknown" and is alerted.	Alert $\leq 30 \text{ s}$; count never negative	Sensor failure	Fault-injection test
NFR-REL-04	Reliability (recoverability)	Recover the last consistent state after power/crash; open sessions preserved.	RTO $\leq 2 \text{ min}$, RPO $\leq 30 \text{ s}$	Crash/power loss	Recovery test
NFR-SEC-01	Security (confidentiality)	Card IDs and payment data encrypted at rest and in transit; no full card/payment data in logs.	AES-256 at rest, TLS 1.2+ in transit	Always	Config review, pen-test
NFR-SEC-02	Security (integrity/authorization)	Every privileged operation is authorized by RBAC and object-level checks.	0 unauthorized actions in test suite	Always	Authorization test

ID	25010 characteristic	Requirement	Metric / target	Condition / load	Verification
NFR-SEC-03	Security (accountability)	100 % of entry/exit/payment/admin events logged, immutable, retained ≥ 12 months, queryable.	100 % coverage; retention ≥ 12 mo	Always	Audit-log review
NFR-SEC-04	Security (non-repudiation)	Detect anti-passback, clone-card, and plate-mismatch conditions.	Alarm raised on each condition	Always	Scenario test
NFR-SCAL-01	Performance efficiency (scalability)	Scale from a 200-slot lot to $\geq 5,000$ slots / multiple sites with no redesign; adding a lot is configuration only.	$\geq 5,000$ slots, config-only	Growth	Architecture review + config test
NFR-USE-01	Interaction capability (usability)	Entrance sign readable at distance; first-time driver interprets indicators; operator finds a vehicle fast.	Readable ≥ 30 m daylight; ≥ 95 % interpret unaided; find-by-plate ≤ 15 s	Field	Usability test
NFR-SAFE-01	Safety	The barrier must be fail-safe : it shall not close on a vehicle or person and shall warn on hazard.	0 unsafe closures	Vehicle/person present	Safety test / interlock review
NFR-MAINT-01	Maintainability / Flexibility	Services are modular and independently deployable; a sensor vendor is replaceable behind an abstraction.	Sensor swap = adapter change only	Change request	Code review
NFR-COMP-01	Compatibility	Integrate with HCMUT_SSO (CAS), read-only HCMUT_DATACORE, and BKPay via standard protocols.	Conforms to each external interface	Integration	Integration test

8. Assumptions, Constraints & Open Questions

8.1 Key assumptions (explicitly stated)

1. **HCMUT_SSO is a CAS server** (Apereo/Jasig CAS, self-reported v3.5.1), not OAuth2/OIDC. The system integrates as a CAS service (redirect $\rightarrow$ service ticket $\rightarrow$ `/p3/serviceValidate`) using a

- CAS client library. If graders permit an abstract identity provider, it may be modelled as OIDC.
2. **HCMUT_DATACORE is the read-only system of record.** The system provisions a local user record on first login from SSO attributes, then enriches/refreshes from DATACORE via on-demand cache with a TTL (≈ 24 h for profile). The local copy is a cache; the system **never writes** to DATACORE.
 3. **BKPay has no public parking API** — it is web-only, single-bank (OCB), and tuition/service-fee scoped. For the prototype the payment gateway is **stubbed**, designed to the standard Vietnamese-gateway shape (redirect + server-to-server IPN, HMAC-signed, idempotent). A real BKPay/OCB integration is future work.
 4. **Primary vehicle is the motorbike**, handled in barrier-free multi-lane flow with ANPR + card tap; barriers apply to the car lane only.
 5. **Occupancy is count-based per zone** (increment on entry, decrement on exit) with nightly recalibration to counter drift.
 6. The prototype demonstrates one representative lot; the model supports multi-campus, multi-gate.

8.2 Constraints

- Must integrate with existing HCMUT_SSO, HCMUT_DATACORE, and BKPay.
- Must enforce role-based access control (end user, operator, administrator).
- Must operate under intermittent connectivity and high concurrency.
- All parking and financial activity must be logged for audit.

8.3 Open questions (to confirm with stakeholders)

1. **Which user attributes does `/p3/serviceValidate` release** for a parking service (uid, email, name, **affiliation/role**, faculty)? This decides how much DATACORE data is actually needed.
 2. Does DATACORE expose a query API (for on-demand caching) or only batch export? Does it hold vehicle/plate registration?
 3. Does any BKPay/OCB micropayment or webhook API exist, or must parking use the "service fee" rail? Is a prepaid wallet acceptable to finance?
 4. Will the university grant a CAS service registration and service URL, and what is the single-logout behaviour?
 5. Barrier safety-certification requirements (fail-safe on vehicle presence).
 6. Is Reservation (Module E) in scope for the first release?
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9. Glossary

Term	Meaning
ANPR / LPR	Automatic Number-Plate Recognition — camera-based reading of the licence plate.
CAS	Central Authentication Service — the SSO protocol HCMUT_SSO uses (ticket-based).
BKNetID / MSSV	HCMUT network account / student ID used to log in via SSO.
BKPay	HCMUT's internal payment platform (bank rail: OCB).
DATACORE	HCMUT_DATACORE — read-only master data for identity and vehicle registration.
RBAC / ABAC	Role-Based / Attribute-Based Access Control.
MQTT	Lightweight publish/subscribe messaging protocol used between IoT gateways and the backend.
IPN	Instant Payment Notification — the gateway's server-to-server payment callback (source of truth for settlement).
Anti-passback	A rule preventing the same credential/vehicle from entering twice without an intervening exit.
Session	A ParkingSession — one park event from entry to exit, the unit of billing.
Staleness timeout	The age after which a slot's last-known state is downgraded to "unknown".

